Introduction to Hybrid Manufacturing 151

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Class Outline

Introduction to Hybrid Manufacturing
Additive Versus Subtractive Manufacturing
Hybrid Additive Manufacturing Tools
Common Additive Processes for Hybrid Manufacturing
Review: Hybrid Tools and Processes
Hybrid Additive Manufacturing Workflows
Advantages of Hybrid Additive Manufacturing
Specialized Applications for Hybrid AM Machines
Industry Applications for Hybrid AM
Review: Hybrid AM Applications

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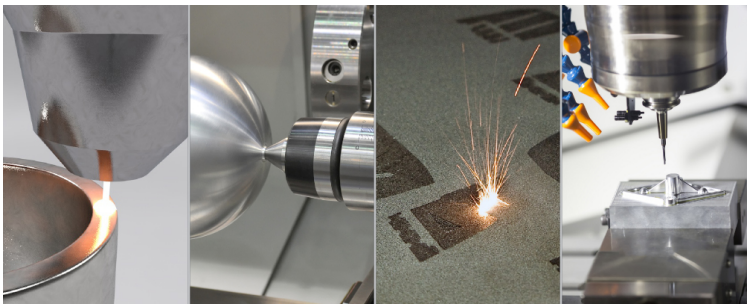
Introduction to Hybrid Manufacturing



Design for both subtractive and additive manufacturing relies on computer-aided design (CAD) software. Additionally, while early subtractive methods could be done manually, today, part production with subtractive and additive methods relies on computer-aided manufacturing (CAM) technology. CAM typically prepares digital instructions for computer numerical control (CNC) software to control tool movements on a machine. These programmed tool movements cut or build a part based on a CAD model. CAD/CAM systems, combined with the enhanced digital connectivity of the Industrial Internet of Things (IIoT), allow manufacturers to streamline design and production workflows and also to combine processes in new ways. Harnessing digital networks to combine separate manufacturing processes into a single workflow is often called hybrid manufacturing. Most often, hybrid manufacturing processes combine additive and subtractive manufacturing methods into one process, which is sometimes called hybrid additive manufacturing (hybrid AM). While subtractive and additive methods have different advantages, both also have certain limitations. Many manufacturers use hybrid additive manufacturing to optimize part design and production in a variety of ways.

This material is based upon research supported by, or in part by, the U.S. Office of Naval Research under award number N00014-18-1-2881 led by the National Center for Defense Manufacturing and Machining (NCDMM).

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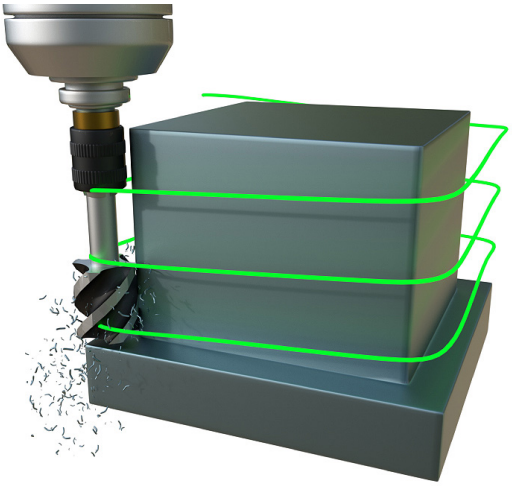
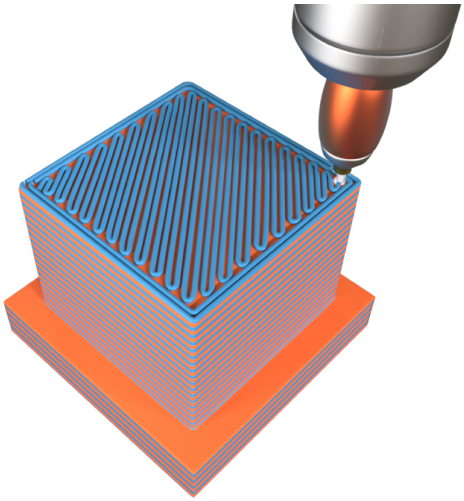
Additive Versus Subtractive Manufacturing

A hybrid additive manufacturing workflow combines the strengths of both additive and subtractive manufacturing methods.



Subtractive Process

A subtractive manufacturing process builds a part by removing material from a workpiece. Typically, designers use CAD software to create a part design, and an engineer uses CAM tools to convert the CAD model into a part program. The part program maps out toolpaths for cutting or finishing tools on a CNC machine. The CNC machine cuts the workpiece into the shape of the finished part based on the CAD dimensions. Subtractive processes can remove very specific amounts of material from a workpiece. This is ideal for making highly precise holes or other part features that must meet very tight tolerances, or for producing a high-quality surface finish. However, subtractive processes often create large amounts of waste as any workpiece material that the process removes becomes scrap. Additionally, CAD models for subtractive methods must account for workpiece and tool dimensions, cutting speed, depth of cut, feed rate, and many other factors.



Additive Process

An additive manufacturing process can build a part from scratch by fusing layers of material, such as melted plastic or metal powder. Like subtractive, most additive parts also begin with a CAD model. However, since additive processes build a single layer of the workpiece at a time, converting from CAD to CAM is easier to automate than with subtractive methods. This can eliminate the need for a skilled engineer to program complex toolpath sequences. AM also makes design customization easier because it is ideal for building complex geometrical shapes. However, since material is deposited in specific quantities, and melted material changes dimensions as it cools, additive methods cannot create precise holes or meet tight tolerances as effectively as subtractive methods. Additionally, AM processes are unable to produce parts with high-quality surface finishes.

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Hybrid Additive Manufacturing Tools

Hybrid additive manufacturing may refer to various production workflows that apply subtractive and additive processes to a single part. However, simply applying different processes to produce a part is not enough to be considered hybrid manufacturing. A hybrid manufacturing process uses a combined digital workflow and also combines multiple processes at a single location.

Some hybrid AM processes may use separate additive and subtractive machines to complete a part, but these machines exchange digital information or share software on a digital network. Networked machines in a hybrid AM process may share CAD/CAM data, as well as data collected and analyzed by smart sensors on each machine in real time. This real-time data may be analyzed throughout the manufacturing process to eliminate unnecessary idle time on each machine and optimize the production workflow.

A hybrid machining center combines both digital and mechanical tools for additive and subtractive processes on a single, all-in-one machine. Like a traditional CNC machining center, a hybrid machining center consists of a large cabinet, a worktable, and an automatic tool changer, but hybrid machining centers can alternate between cutting tools and additive tools. Hybrid machining centers can create unique parts that could not be produced by separate machines since operators can use both tool types at any point without relocating the part. If a manufacturer plans to use both additive and subtractive technology, using a hybrid machining center may be more practical and cost-effective than purchasing CNC machining centers or CNC turning centers and AM machines separately.

However, hybrid manufacturing does not necessarily require the purchase of an additional machine. Many print nozzles are designed to be added to traditional CNC machines as print head attachments in order to convert subtractive machines into hybrid machines. Hybrid machine attachments may be practical for manufacturers looking to integrate additive capabilities into their existing operations.

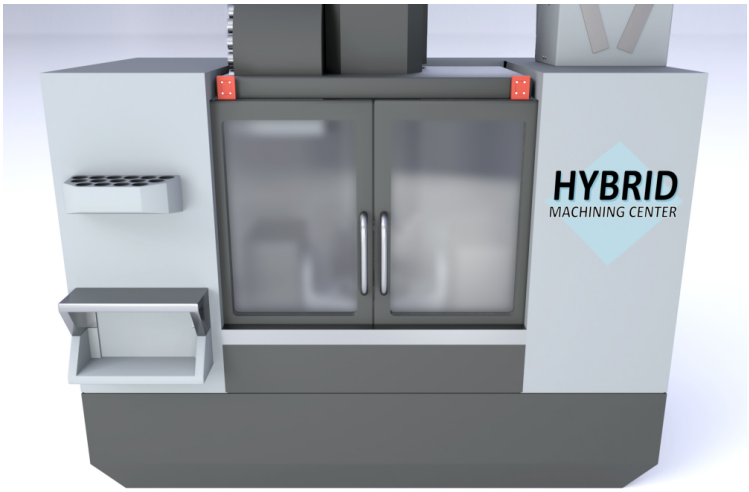

Figure 1: All-In-One Hybrid Machine

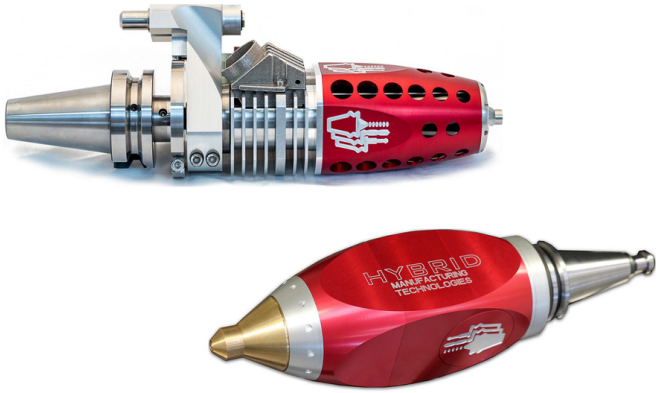

Image courtesy of Hybrid Manufacturing Technologies

Figure 2: Print Head Attachments

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Common Additive Processes for Hybrid Manufacturing

Most all-in-one hybrid machines combine a metal AM process, usually directed energy deposition (DED) or powder bed fusion (PBF), with a material removal process. Directed energy deposition uses a print nozzle to dispense powdered material from a material hopper or wire material from a material spool. The nozzle also emits a laser, arc, or other thermal energy source that melts dispensed material into layers to build a part. In a typical hybrid DED process, once the print nozzle builds the part, cutting tools such as mills, drills, turning inserts, or abrasives are then used to complete the part so it meets design specifications. Powder bed fusion most often uses a laser to heat and fuse layers of powdered metal or other powdered material together to build a part. As each part layer is completed, the build platform supporting the part sinks lower into the powder bed until all layers are complete. In hybrid applications, PBF is combined with machining tools such as drills, end mills, or face mills. Typically, PBF builds one or more layers of the part, then a machining tool trims the finished layers to specifications before the part is lowered and the PBF laser fuses the next layer.

Rather than using DED or PBF, some additive hybrid processes may use ultrasonic additive manufacturing (UAM), a type of sheet lamination. UAM is an additive process that bonds together thin sheets of metal using ultrasonic waves. In UAM, as a sheet of metal is pressed against a metal base or other sheet of metal, ultrasonic vibrations create friction that causes the metal to bond together in a solid state. In hybrid UAM applications, once the metal is bonded, a cutting tool cuts the metal layers to design specifications. While UAM can be ideal for solid-state bonding of multiple metals, it does not typically provide the same level of complex geometry as hybrid DED and PBF methods.

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Review: Hybrid Tools and Processes

1. Hybrid sheet lamination processes can easily create more complex part dimensions than hybrid directed energy deposition (DED) processes. **FALSE**
2. Directed energy deposition (DED) is one of the most common additive manufacturing methods used in hybrid manufacturing processes. **TRUE**
3. A hybrid machining center can eliminate the need for separate additive and subtractive manufacturing machines. **TRUE**
4. Hybrid manufacturing processes provide less precision and accuracy than traditional subtractive manufacturing methods alone. **FALSE**
5. Hybrid all-in-one machines can produce components that would be impossible to produce using separate additive or subtractive machines. **TRUE**

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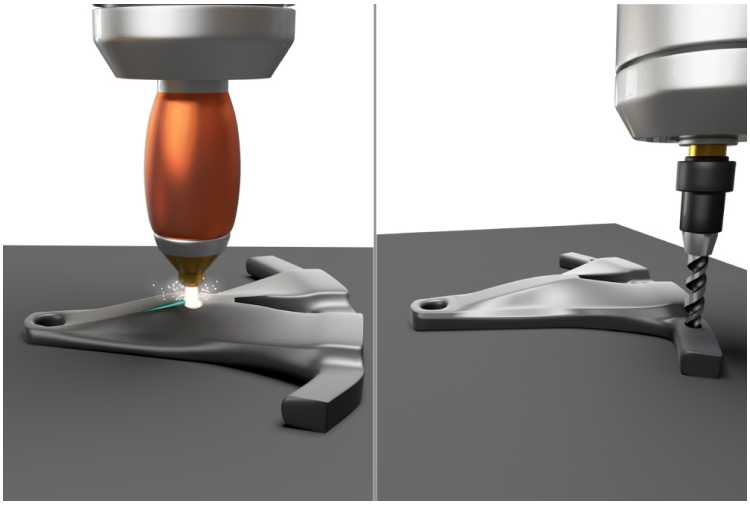
Hybrid Additive Manufacturing Workflows



Additive-to-Subtractive Workflow

In most hybrid AM production workflows, a machine builds a near-net shape workpiece using an additive process, such as DED. Next, once the workpiece is built, a subtractive process, such as CNC machining completes the part. However, additive and subtractive processes may also be alternated throughout the build process to create unique part features.

Forming-to-Additive Workflow

Alternatively, rather than building the entire workpiece with additive methods, a hybrid AM workflow may start with a forming process to create a casting or forging, then use an additive process to complete the part. Castings and forgings are net shape or near-net shape parts created by a mold. Since additive processes can be time consuming, using castings or forgings for the part base can greatly reduce production times. AM tools can then apply additional materials to create final part features or apply high-performance, and often expensive, materials in specific areas to optimize performance and save costs. Typically, this hybrid AM workflow also uses a finishing process to improve surface quality of the completed part.

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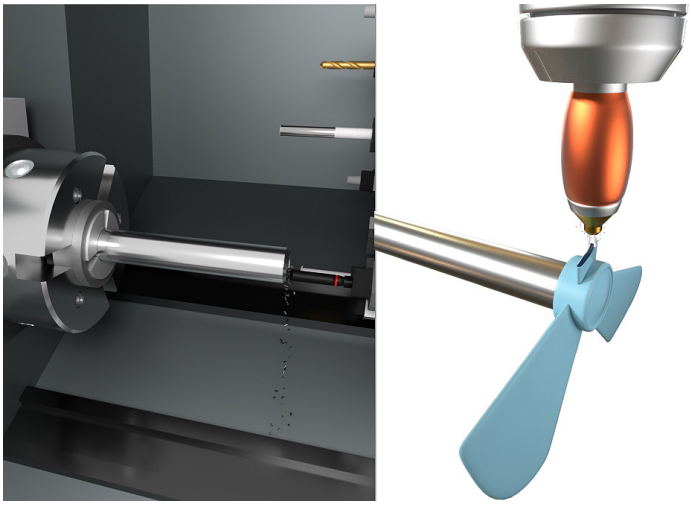

Subtractive-to-Additive Workflow

Hybrid AM processes may also begin with subtractive methods on a billet of material, then use additive methods to complete the part. For example, a marine manufacturer may use cutting tools to machine a boat propeller shaft to specified dimensions, then complete the propeller by building the blades on the shaft with a DED nozzle or other AM tool.

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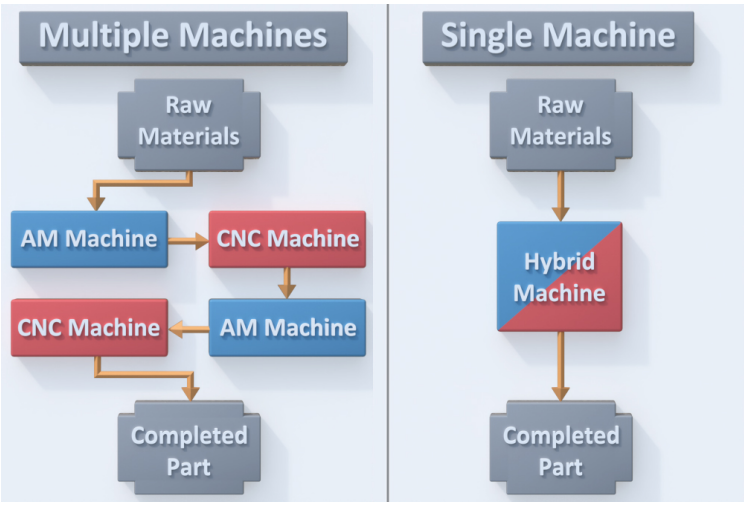
Advantages of Hybrid Additive Manufacturing

The overall advantage of hybrid AM is that it combines the design complexity and reduced material costs of AM with the superiour accuracy and surface finish of subtractive methods. Material costs can be reduced using an all-in-one hybrid machine or using separate additive and subtractive machines, but using separate machines can slow production since parts and materials must be transferred between steps. A hybrid AM machine, however, does not have this disadvantage and greatly improves production efficiency. All-in-one hybrid machines and hybrid AM machine attachments can have high up-front costs and may require additional training for personnel. However, manufacturers often consider the total cost of ownership (TCO) when comparing hybrid AM machines to traditional production methods. The TCO includes equipment, labor, time, and other costs related to owning particular machines or equipment. A hybrid AM machine reduces TCO over the long term.

In addition to reducing production times, hybrid AM machines allows manufacturers to automate the additional inspection steps often required when using AM methods. AM typically requires multiple inspection steps throughout the build process to adjust for defects or inconsistencies in the build. For example, when creating a near-net shape part with AM, technicians may need to inspect the part before it is completed to verify that part dimensions are accurate based on the CAD model. Automating inspection on a hybrid AM machine using probes, laser scanners, or other methods optimizes inspection times, significantly improves quality, and reduces part defects.



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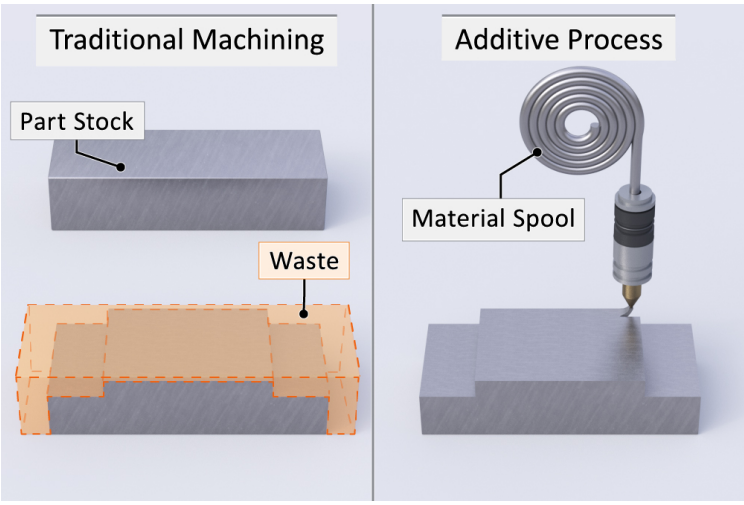

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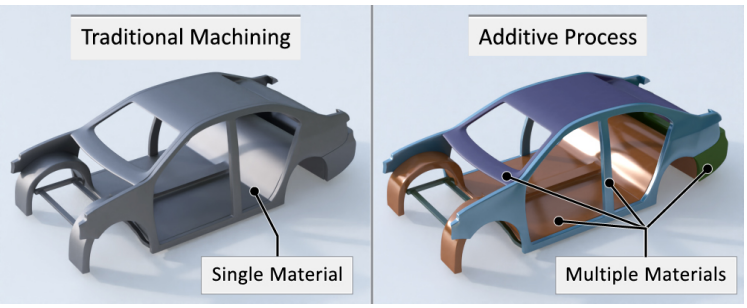
Specialized Applications for Hybrid AM Machines



Alternating between additive and subtractive processes on one machine allows manufacturers to produce unique part features while also improving the precision of each feature. Additionally, hybrid AM machines can finish deep or internal part features, such as internal holes or pockets, during the build process before external part features block access to these surfaces. AM tools on hybrid AM machines also allow manufacturers to design and build components made up of different layered materials. For example, adding a layer of copper improves electrical conductivity, and adding small quantities of high-performance metals, such as superalloys, improves hardness or other properties. These AM tools can greatly reduce material costs since expensive materials are only used where needed. Combining AM tools with cutting and finishing tools allows hybrid AM machines to produce highly accurate parts with unique properties at a lower cost than traditional methods.

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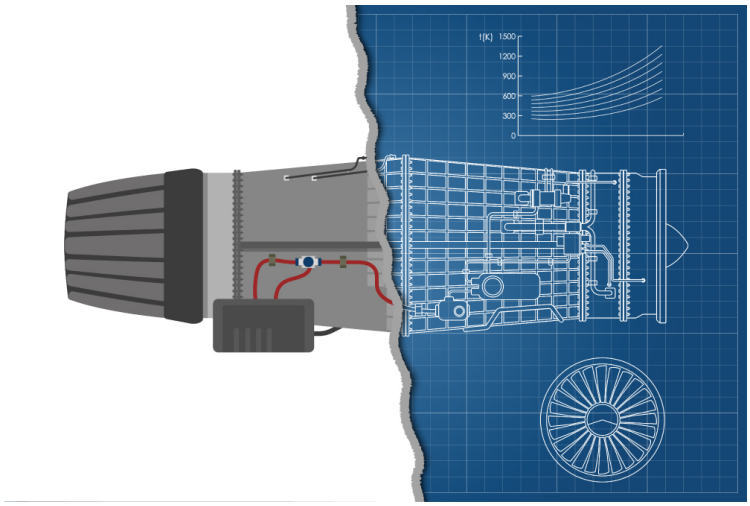
Industry Applications for Hybrid AM

Dental Implant

Hybrid AM processes can produce innovative new devices that are highly customized. For example, a manufacturer can use AM to produce complex parts like prosthetic limbs or dental implants, then machine the parts down to highly precise, personalized dimensions. This can improve functionality and overall performance of the medical device.



Jet Engine

Equipment for the aerospace industry often requires highly customized parts and components. Aerospace parts are typically produced in small lots and must meet very high tolerances. Hybrid AM processes allow manufacturers to produce these very precise custom parts within required tolerances. These parts often require superalloys and other expensive metals, and hybrid AM processes help reduce material costs by reducing waste. This can help aerospace part manufacturers meet high quality standards while improving lean manufacturing and other waste management practices.

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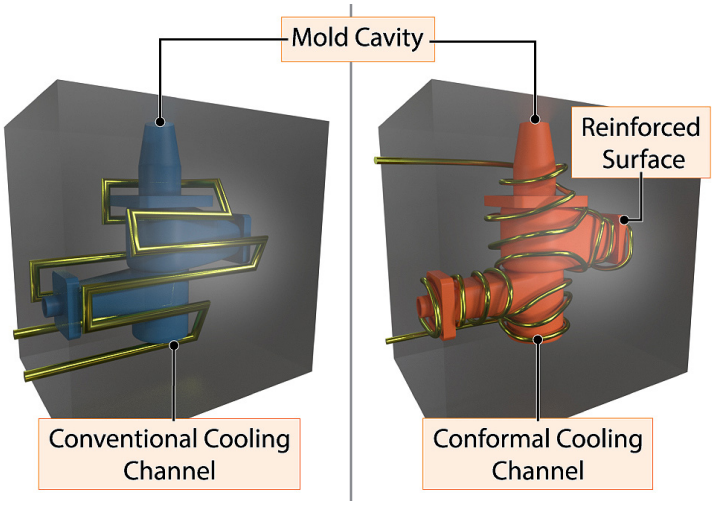
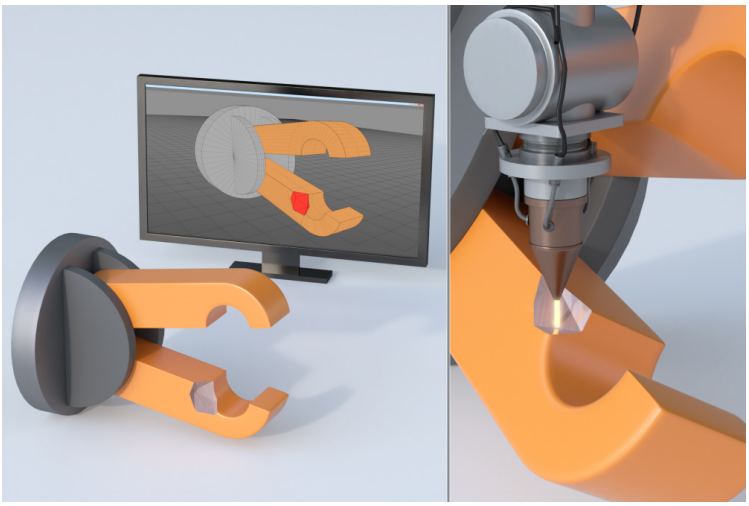


Precision Mold with Conformal Cooling

Traditional AM methods can produce customized tooling, such as custom robot grippers or a custom mold for a part casting. With AM, manufacturers can design custom molds that use conformal cooling channels rather than conventional channels made with subtractive processes. Hybrid AM allows manufacturers to produce custom molds with conformal cooling that are also highly precise. Producing custom molds on a hybrid AM machine can improve repeatability and increase production volume. In addition, hybrid AM machines can also reinforce the mold cavity surface with a single layer of specified metal or other material that may help the part casting cool and solidify faster or provide other benefits.



DED Repair Process

Repair and maintenance operations can also combine custom molding with both additive and subtractive processes to repair damaged parts and components more accurately. In many hybrid AM repair processes, the manufacturer creates a mold of the damaged part, then uses a DED nozzle to rebuild the part component from the mold. Once the replacement part component is built, subtractive processes, such as grinding, may bring the replacement component to final quality specifications.

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Review: Hybrid AM Applications

Which of the following can create a net-shape part in a hybrid manufacturing process?

- **Molding**
- Stamping
- Bending

Which of the following is often required to ensure quality between stages of a hybrid additive manufacturing process?

- **Integrated inspection steps.**
- Customized molds or dies.
- Forging and casting processes.

Hybrid manufacturing is most effective for which of the following?

- **Finishing deep or internal part features during production.**
- Mass producing simple parts at very high volumes.
- Integrating milling and turning tools in a single process.

Which type of application can be improved with hybrid additive manufacturing?

- **Mass customization**
- Machine learning
- Cloud computing
- Customer segmentation

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Class Vocabulary

Abrasives A material consisting of hard particles used to remove workpiece material. Abrasives are used in both heavy and light material removal.

Additive Manufacturing AM. The process of joining or solidifying materials to make an object from a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object.

Aerospace The industry concerned with the research, design, manufacture, and operation of air and space craft. Aerospace products include airplanes, helicopters, rockets, missiles, satellites, space capsules, space planes, and related systems.

AM Additive manufacturing. The process of joining or solidifying materials to make an object from a three-dimensional computer model. AM methods typically build up layers of material to create an object.

Arc An area in which electricity jumps from an electrode on a tool to a workpiece. An arc melts the powdered metal or wire feedstock in directed energy deposition.

Automatic Tool Changer A CNC mill or machining center component that holds and changes tools during operation. Automatic tool changers reduce cycle times by automatically changing tools between functions.

Batches A specific number of the same part that moves through the production cycle. Small batch manufacturers produce a variety of different products but in low volume.

Billet A small, semi-finished piece of metal that is rectangular, circular, or square in shape. Billets are used in semi-solid molding.

Build Platform The flat surface on which a part is additively manufactured. The build platform can either be a permanent machine surface from which parts are removed or a surface that can be removed from a machine once the build is complete.

CAD Computer-aided design. A method of designing two- and three-dimensional objects using computers and software. CAD is most often used to create part models for production.

CAM Computer-aided manufacturing. The use of computer software that facilitates the development of part programs to produce a part. CAM software applications create an image of the workpiece and develop the program code from information that the programmer inputs.

Casting A workpiece formed by pouring molten metal into a mold and cooling it into a solid shape. Castings are near their finished shape directly after they are formed.

Computer Numerical Control CNC. A programmable system of software and hardware that directs the operation of a machine. Computer numerical control systems use mathematical data to direct machine movements.

Computer-Aided Design CAD. A method of designing two- and three-dimensional objects using computers and software. Computer-aided design is most often used to create part models for production.

Computer-Aided Manufacturing CAM. The use of computer software that facilitates the development of part programs to produce a part. Computer-aided manufacturing software applications create an image of the workpiece and develop the program code from information that the programmer inputs.

Conformal Cooling Cooling channel configurations that curve and closely correspond to a part's shape. Conformal cooling provides manufacturers with better heat management and allows them to reduce tool wear and cycle time.

Customization Manufacturing a part with unique design variations. Customization is typically based on specific requests from customers.

DED Directed energy deposition. An additive manufacturing process in which focused thermal energy is used to melt materials as they are deposited. DED is often used with powdered or wire metal feedstock.

Design Specifications A detailed plan for a part or object that includes dimensions and other precise descriptions of its manufacturing requirements. Design specifications are typically created using computer-aided design (CAD) software.

Digital Consisting of information that is input or output electronically as a series of pulses or signals, often resulting in binary strings of 0s and 1s. Digital computing devices interpret various programming commands as binary digits.

Directed Energy Deposition DED. An additive manufacturing process in which focused thermal energy is used to melt materials as they are deposited. Directed energy deposition is often used with powdered or wire metal feedstock.

Drills A multi-point cutting tool used to create a round hole in a workpiece surface. Drills are common tools for holemaking operations on a CNC mill or lathe.

Electrical Conductivity The ability of a material to act as a medium for conveying electricity. Metals such as copper and aluminum have high levels of electrical conductivity.

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Features A defining characteristic on a component or part. Features include corners, edges, holes, and grooves.

Forging A workpiece formed by compressing metal between two dies to achieve a specific shape. Forging operations typically heat the metal to an elevated temperature.

Forming A manufacturing process that forms parts by shearing, stretching, bending, or compressing metal or other material. Forming processes do not create chips like subtractive processes.

Grippers A mechanical device on a robot that uses rotary and linear actuators to mimic the motion of the human hand. Grippers have complex shapes that can be easily produced using additive manufacturing.

Hybrid Additive Manufacturing Hybrid AM. A process that uses both traditional and additive manufacturing (AM) to create a finished part. Hybrid additive manufacturing involves combining additive and subtractive processes on a single machine.

Hybrid AM Hybrid additive manufacturing. A process that uses both traditional and additive manufacturing (AM) to create a finished part. Hybrid AM typically involves combining additive and subtractive processes on a single machine.

Hybrid Machining Center A type of CNC machine that is capable of both additive manufacturing and traditional subtractive machining. Hybrid machining centers typically resemble the basic structure of a traditional CNC mill.

Hybrid Manufacturing An manufacturing approach that produces parts by combining multiple manufacturing processes in a single digital workflow or on a single machine. Hybrid manufacturing may describe a variety of manufacturing processes combinations but is most often associated with combining additive and subtractive processes.

Industrial Internet Of Things IIoT. A network of physical devices used in manufacturing that contain computing systems that allow them to send and receive data. The Industrial Internet of Things allows devices to exchange data and automate processes without any human intervention.

Laser An intense beam of light that can be precisely aimed and controlled. Lasers are sometimes used to selectively solidify or fuse materials in additive manufacturing methods, including directed energy deposition, powder bed fusion, and vat photopolymerization.

Lean Manufacturing An approach to manufacturing that seeks to reduce the cycle time of processes, increase flexibility, and improve quality. Lean manufacturing approaches help to eliminate waste in all its forms.

Lots The number of parts made with a particular tooling setup. A lot is composed of identical or similar parts or materials.

Machining A manufacturing process that uses cutting tools to remove material to form an object. Machining includes methods such as milling, turning, and drilling.

Mass Customization Manufacturing a large number of uniquely designed variations on a part. Mass customization is a key advantage of additive manufacturing.

Material Hopper A container used to hold raw materials for use in an industrial process. Material hoppers used in metal AM processes hold powdered metal.

Material Removal The process of removing substances or matter from a surface. Material removal processes include cutting, grinding, deburring, deflashing, polishing, waterjet cutting, and routering.

Material Spool A section of raw material in thin wire form wrapped tightly around a base. Material spools are attached to additive print nozzles so that the raw material in the spool can be dispensed.

Mills A rotating, multi-point cutting tool that is guided along a workpiece to create flat surfaces, slots, or other features. The term mill also refers to the machine that uses mills to perform milling operations on a workpiece.

Mold A hollow cavity used to shape material. In metal casting, the mold is filled with liquid metal, which takes the shape of the mold as it solidifies.

Near-Net Shape A part with dimensions that are close to its final specified form after initial manufacturing steps. Near-net shape parts require little subsequent machining.

Net Shape A part that meets consumer or manufacturer specifications, including tolerance and surface finish, directly after it is manufactured. Net shape parts do not need any further post processing after they are created.

Part Program A series of instructions produced by CAM software and used by a CNC machine to perform the necessary sequence of operations to complete a part based on a CAD design. Part programs are typically written in G code.

Powder Bed Fusion PBF. An additive manufacturing method that uses thermal energy to melt together layers of powdered polymer, metal, ceramic, or other material. Powder bed fusion processes often use either lasers or electron beams as thermal energy sources.

Powdered Metal A metallic powder that is often used as a build material in additive manufacturing (AM). Powdered metal is stored in a material hopper and deposited through a nozzle or print head to build a part layer by layer in AM applications.

Print Head Attachments A portable additive manufacturing component that holds and distributes part build material. Print head attachments can be installed on many traditional CNC machines to convert them into hybrid machines.

Raw Materials An unprocessed or lightly processed substance, often referred to as feedstock, that is used in additive manufacturing (AM) machines to build parts or components. Raw materials for AM commonly come in powder,

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pellet, wire, or semi-liquid form.

Real Time The near-instantaneous interval of time that computers require to process data. Real time is virtually the same as actual time because computers process data nearly immediately.

Repeatability The ability of a machine or process to continually deliver accurate and precise results. Repeatability measures how consistently a machine can replicate identical or near-identical parts.

Sheet Lamination An additive manufacturing process that forms an object by bonding sheets of material together using an adhesive, heat, brazing, or ultrasonic waves. Sheet lamination processes are sometimes ideal for creating prototypes but are not commonly used to manufacture multiple finished products.

Smart Sensors A device equipped with software that can detect physical inputs, process them as data, and output digital signals. Smart sensors are more advanced than normal digital sensors since they can process data internally rather than simply sending digital signals to an external system to be processed.

Subtractive Manufacturing Any manufacturing process in which a piece of raw material is machined into a desired final shape through a controlled material removal process. Subtractive manufacturing methods are fast but create a large amount of waste.

Superalloys An expensive, complex metal alloy designed to perform under intense conditions, such as elevated temperatures. Superalloys are also known as high performance alloys.

Tolerances An acceptable deviation from a desired dimension that meets specifications. Parts outside of a required tolerance must be discarded or reworked.

Toolpaths A series of coordinate positions that determine the movement of a tool during a machining operation. Toolpaths are expressed using G code commands.

Total Cost Of Ownership TCO. An estimate of all the direct and indirect expenses associated with a purchase. The total cost of ownership of a hybrid AM machine takes into account several factors, including initial costs, maintenance and repair costs, labor, and the cost of raw materials used to produce parts on the machine.

Turning Inserts An insert designed for turning operations on a CNC lathe. A turning insert on a hybrid manufacturing machine may be used to remove material from a part feature created with an additive tool.

Ultrasonic A frequency above the range of human hearing. Ultrasonic frequencies are used to bond sheets of metal in ultrasonic additive manufacturing (UAM).

Ultrasonic Additive Manufacturing UAM. A sheet lamination process that uses high-frequency vibrations to bond successive layers of material together to build a final part. Ultrasonic additive manufacturing most frequently uses sheets of metal foil or tape as build material.

Workpiece A material being machined or undergoing another type of processing. A workpiece may be subject to cutting, welding, forming, additive printing, or other operations.

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